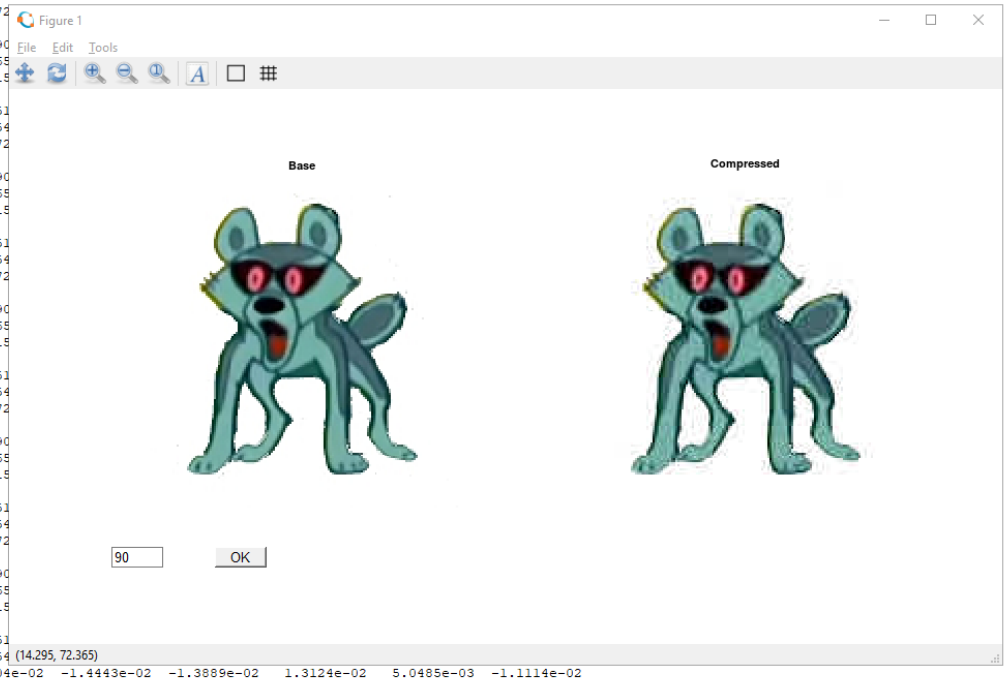
Андреев Дмитрий Павлович 431-3

```
-1.5790e-16   3.8065e-16            0            0   1.1975e-16   1.4211e-16
 3.8065e-16  -2.1210e-16  -9.1095e-17   1.1975e-16  -3.5527e-17   3.2006e-17
-3.1815e-16            0  -1.5967e-16  -3.5527e-17  -3.2006e-17  -7.2802e-17
          0   3.9918e-17  -3.5527e-17  -6.4013e-17  -1.4560e-17  -6.6780e-18
-2.3951e-16   3.5527e-17  -6.4013e-17   2.0384e-16   1.3356e-17  -1.1102e-16
 7.1054e-17  -1.6003e-17  -1.6016e-16   6.0102e-17   6.1679e-18   4.2976e-17
-7.4272
```



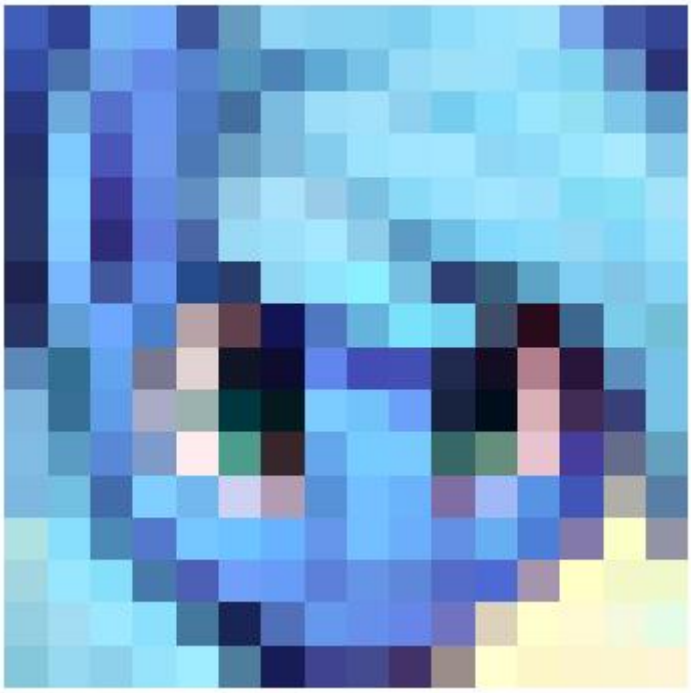

```
 3.5504e-02  -1.4443e-02  -1.3889e-02   1.3124e-02   5.0485e-03  -1.1114e-02
-3.3314e-02   1.5136e-02   1.5480e-02  -1.5214e-02  -6.0149e-03   1.3510e-02
 2.3710e-02  -1.1457e-02  -1.2188e-02   1.2311e-02   4.9656e-03  -1.1327e-02
-1.7147e-02   8.6183e-03   9.4221e-03  -9.7099e-03  -3.9774e-03   9.1835e-03
 1.2188e-02  -6.2956e-03  -7.0225e-03   7.3492e-03   3.0473e-03  -7.1057e-03
-8.2256e-03   4.3351e-03   4.9106e-03  -5.2020e-03  -2.1784e-03   5.1212e-03
```

```
%Определяем разрешение экрана
ScreenSize = get(0,'ScreenSize');
handles.figure = figure('Position',[ScreenSize(3)/4,ScreenSize(4)/4,ScreenSize(3)/2,ScreenSize(4)/2]);

handles.E=uicontrol('Style','edit','Position',[100,50,50,20],'String', '90');
set(handles.E,'BackgroundColor',[1 1 1]);
set(handles.E,'HorizontalAlignment','left');
handles.B=uicontrol('Style','pushbutton', 'Position',[200,50,50,20], 'string', 'OK', 'Callback', 'Decode()');

A = imread('C:\Users\kev4313\Desktop\laba7\jackal.png');
handles.A = A;
guidata(handles.figure, handles);

subplot(1,2,1);
imshow(A);
title('Base')



function Decode()
  handles=guidata(gcf);
  strParamVal = get(handles.E, 'String');
  disp(str2num(strParamVal));
  R = handles.A(:,:,1);
  G = handles.A(:,:,2);
  B = handles.A(:,:,3);
  strParamVal=str2double(strParamVal);


  Y=0.299*double(R)+0.587*double(G)+0.114*double(B);
  U=-0.14713*double(R)-0.28886*double(G)+0.436*double(B)+128;
  V=0.615*double(R)-0.51499*double(G)-0.10001*double(B)+128;

  DY = blockproc(Y,[8 8],@mydct);
  disp("DY = blockproc(Y,[8 8],@mydct)");
  DU = blockproc(U,[8 8],@mydct);
  disp("DU = blockproc(U,[8 8],@mydct)");
  DV = blockproc(V,[8 8],@mydct);
  disp("DV = blockproc(V,[8 8],@mydct)");
  disp(DY);
  MT=zeros(8,8);
  for i = 0:1:7
    for j = 0:1:7
      MT(i+1,j+1)=1+(i+j)*(100-strParamVal+1);
```

```
    endfor
  endfor
  disp(MT);
  DY=blockproc(DY,[8 8],@knv,MT);
  DV=blockproc(DV,[8 8],@knv,MT);
  DU=blockproc(DU,[8 8],@knv,MT);
  disp(DY);
  DY= round(DY);
  DU= round(DU);
  DV= round(DV);

  DY=blockproc(DY,[8 8],@iknv,MT);
  DV=blockproc(DV,[8 8],@iknv,MT);;
  DU=blockproc(DU,[8 8],@iknv,MT);

  DY = blockproc(DY,[8 8],@myidct);
  DU = blockproc(DU,[8 8],@myidct);
  DV = blockproc(DV,[8 8],@myidct);

  B(:,:,1)=DY+1.13983*(DV-128);
  B(:,:,2)=DY-0.39465*(DU-128)-0.58060*(DV-128);
  B(:,:,3)=DY+2.03211*(DU-128);
  B=uint8(B);
  subplot(1,2,2);
  imshow(B);
  title('Compressed')
endfunction
```

```
A=imread('C:/Users/kev4313/Desktop/Lab7_1/AJ.png');
A_R=A(:,:,3);
A_G=A(:,:,2);
A_B=A(:,:,1);
subplot(2,2,1);
imshow(A) ;
title ('RGB');
subplot(2,2,2); #значит, что создается изображение 2x2, текущий
imshow(A_B);
title('Blue channel');
subplot (2,2,3); #значит, что создается изображение 2x2, текущий
imshow(A_G) ;
title('Green channel');
subplot(2,2,4); #значит, что создается изображение 2x2, текущий
imshow(A_R);
title('Red channel');
saveas (gcf, 'MyFigure.png');
figure;
comb_lily(:,:,1)=A_R; comb_lily(:,:,2)=A_G; comb_lily(:,:,3)=A_B;
comb_lily=uint8(comb_lily);
imshow(comb_lily);
title('lily combined from separated channels');
```

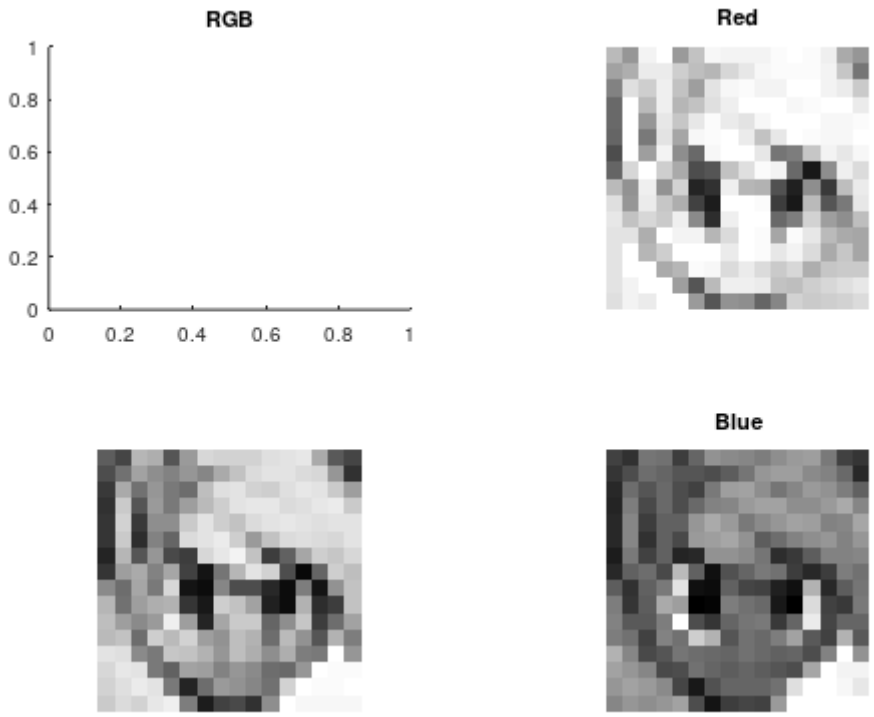
lily combined from separated channels



RGB

Blue channel

Green channel

Red channel

```
figure;
A = imread('C:\Users\kev4313\Desktop\Lab7_1\AJ.png');
imshow(A);


R = A(:,:,1); G = A(:,:,2); B = A(:,:,3);
subplot (2,2,1);
title ('RGB') ;
subplot (2,2,2);
imshow (R) ;
title ('Red');
subplot (2,2,3) ;
imshow (G) ;

subplot (2,2,4) ;
imshow (B) ;
title ('Blue');
Y = 0.299*double(R) + 0.587*double(G) + 0.114*double(B);
U = -0.14713*double(R) - 0.28886*double(G) + 0.436*double(B) + 128;
V = 0.615*double(R) - 0.51499*double(G) - 0.10001*double(B) + 128;
S = size(Y); N = S(1); M = S(2);

DY=int16(DY) ;

DU=int16(DU) ;
DV=int16 (DV) ;

R = Y + 1.13983* (V - 128);
G = Y - 0.39465* (U - 128) - 0.58060* (V - 128);
B = Y + 2.03211* (U - 128);
figure;
NA(:,:,1)=R; NA(:,:,2)=G; NA(:,:,3)=B;
NA=uint8 (NA) ;
imshow (NA) ;
title ('New AJ');
```